Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Surface-Modified Upcycling of Spent NMC Cathodes for Cobalt-Reduced Layered Oxides**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical/Electronics			
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	*Secondary Batteries*		
Concept of Proposed Technology	**Core Concept** The proposed technology aims to upcycle end-of-life Nickel–Manganese–Cobalt (NMC) cathode materials into cobalt-lean, performance-enhanced layered oxide cathodes through selective surface compositional modification. Rather than extracting metals or fully resynthesizing cathode materials, this approach treats spent NMC particles as structurally preserved templates and selectively modifies their near-surface composition to improve electrochemical stability and durability. The core concept is the creation of a manganese-enriched surface region on spent NMC particles while largely preserving the original bulk composition and morphology. This surface modification is designed to stabilize the layered lattice, suppress surface degradation phenomena, and improve interfacial compatibility during electrochemical cycling. Subsequent compositional equilibration and lattice reordering steps restore lithium content and structural coherence, yielding a modified layered oxide architecture that retains high energy density while exhibiting enhanced stability.			

	Key Functions and Potential Applications By focusing on surface-level functional enhancement rather than bulk compositional transformation or elemental recovery, the proposed technology avoids energy-intensive processing and minimizes material loss. The approach is broadly applicable across different NMC compositions, making it suitable for heterogeneous spent cathode streams while enabling meaningful performance upgrading without full cathode resynthesis.
Rationale for Joint R&D Project	**Critical Technical Barriers** Upcycling spent NMC cathodes through controlled surface compositional modification requires close coordination between materials science, feedstock variability management, and scale-relevant validation. A key challenge is that end-of-life cathode materials enter recycling streams in heterogeneous forms, including recovered cathode powders and black mass, with varying compositions, surface conditions, and aging histories. Therefore, developing a robust upcycling framework demands access to representative feedstocks and validation under realistic material variability, in addition to laboratory-scale materials expertise. **Why Global Collaboration?** Overcoming the scientific and technological barriers associated with upcycling end-of-life battery cathode materials and developing physics-informed multiscale health estimation requires deep interdisciplinary research in surface chemistry, electrochemical stability, materials degradation, and predictive modeling. Purdue University's Green Energy Materials & Systems (GEMS) Laboratory is a globally recognized research group dedicated to advancing sustainable energy storage materials and technologies, including material reuse and recycling strategies for cleaner, more durable battery systems. The GEMS Laboratory integrates expertise in advanced materials characterization, in-operando diagnostic techniques, and multi-physics modeling to understand battery material properties and interactions from the nanoscale to the system level. Such capabilities are directly aligned with the core challenges of selective surface modification and multiscale digital twin development for heterogeneous cathode feedstocks. In addition, Purdue's broader energy research ecosystem—including interdisciplinary collaborations within the Institute for Energy Innovation (IEI), which unites efforts across materials science, energy storage, and computational analysis—further strengthens its capability to support fundamental research and systemic innovation. A joint R&D framework is essential to integrate fundamental understanding of cathode surface chemistry and structural stability with practical access to real recycling-derived feedstocks, including black mass and cathode materials recovered from industrial processes. Purdue provides the capability to systematically evaluate structure–property relationships and electrochemical behavior of surface-modified cathodes, while industry partners contribute critical resources in the form of representative feedstock supply and scale-relevant validation environments. Collaboration ensures that the developed technology is not limited to idealized materials but is applicable to real-world recycling streams and industrial constraints.

Key Words	1. Cathode upcycling 2. Direct recycling 3. Cobalt-lean NMC 4. Surface compositional engineering 5. Circular economy

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2)	End of the Project	Level (5)

R&D Objectives

R&D Objectives

The objective of this R&D project is to develop a framework-preserving surface modification strategy that enhances the electrochemical stability and performance of spent NMC cathodes while reducing functional reliance on cobalt. The project aims to establish reproducible processing conditions that yield manganese-enriched surface regions capable of stabilizing the layered lattice and mitigating degradation during high-voltage cycling.

Detailed Objectives

Additional objectives include restoring lithium stoichiometry following surface modification, maintaining compatibility with conventional electrode fabrication processes, and demonstrating stable electrochemical performance at the cell level. The overarching goal is to validate a scalable and economically viable upcycling pathway that improves the functional value of spent NMC materials without disclosing proprietary implementation details.

R&D Target Specifications

The upcycled NMC cathode materials are targeted to retain a layered oxide structure with a manganese-enriched near-surface region and preserved bulk composition. The materials are designed to achieve a reversible discharge capacity of ≥225 mAh g^{-1} when cycled between 2.8–4.5 V at 25 °C, with ≥90% capacity retention after 300 cycles and ≥85% retention after 500 cycles at C/2. Average Coulombic efficiency after initial conditioning is targeted to be ≥99.7%.

Surface modification effectiveness will be quantified by a ≥1.5× enrichment of Mn concentration at the particle surface relative to the bulk, while maintaining bulk transition-metal composition within ±3 mol% of the starting NMC material. The layered structure will be retained with ≤5 wt% non-layered phases detected. The upcycled powders are targeted to exhibit a tap density of ≥1.8 g cm^{-3} and a median particle size D50 of 6–12 μm. The process is designed to achieve a ≥90% mass yield from the cathode-active fraction and to demonstrate reproducible performance across multiple industrial feedstock batches.

Parameters	Goal
Reversible discharge capacity	≥225 mAh g^{-1} when cycled between 2.8–4.5 V at 25 °C
Capacity retention after 300 cycles	≥90%
Capacity retention after 500 cycles	≥85%
Average Coulombic efficiency after initial conditioning	≥99.7%
Surface modification effectiveness	≥1.5× enrichment of Mn concentration at the particle surface relative to the bulk
Tap density of upcycled powders	≥1.8 g cm^{-3}

Median particle size of upcycled powders	D50 of 6–12 μm

R&D Plan

The project will be executed through a staged development framework that progressively advances from materials design to cell-level validation.

- Stage 1

In the initial stage, spent NMC cathodes will be characterized to establish baseline structural, compositional, and electrochemical properties. Surface modification approaches capable of introducing controlled compositional changes without disrupting particle integrity will then be evaluated using laboratory-scale processing methods. Emphasis will be placed on reproducibility, compositional control, and preservation of the layered lattice.

- Stage 2

In the second stage, post-modification equilibration and stabilization treatments will be optimized to restore lithium balance and ensure structural coherence. Relationships between surface modification, structural stability, and electrochemical behavior will be established through systematic testing under application-relevant conditions.

- Stage 3

In the final stage, the optimized upcycling framework will be adapted to scalable processing environments in collaboration with industry partners. The upcycled materials will be incorporated into electrodes and evaluated in commercial-format cells to demonstrate cycling stability, rate capability, and thermal robustness. This staged approach ensures technical validation while limiting disclosure of proprietary processing details.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The Purdue team will lead the scientific design and validation of the surface gradient-doping strategy. Responsibilities include development of deposition and diffusion protocols, characterization of spatial compositional gradients using surface- and depth-sensitive techniques, and evaluation of electrochemical performance at the material and electrode levels. The institute will also provide a mechanistic understanding of lattice stabilization and surface reconstruction suppression, supporting process optimization and intellectual-property development.
Expected R&R of Korean SME Partner	The Korean SME partner will contribute to process adaptation, scale-relevant validation, and electrode and cell fabrication using the upcycled cathode materials. Their responsibilities include evaluating manufacturability, conducting performance benchmarking in commercial-format cells, and assessing cost and scalability considerations relevant to industrial deployment. Collaboration with the research institute will ensure that validation activities align with real-world requirements while preserving proprietary process knowledge.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The project will deliver a validated upcycling framework that enhances the functional value of spent NMC cathodes through surface-level compositional engineering rather than elemental extraction or full resynthesis. The technology supports reduced cobalt reliance, improved electrochemical stability, and compatibility with existing manufacturing infrastructure. By demonstrating that spent cathode materials can be systematically upgraded without disclosure of sensitive processing details, the project contributes to sustainable battery supply chains and establishes a foundation for future commercialization and intellectual property development.